

VINAY RAJU SANNIBOINA

Graduate Automotive & Electric Vehicle Engineer

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PROFESSIONAL PROFILE

MSc Automotive Engineering (Electric Vehicles) graduate with a First Class BEng in Electrical & Electronics Engineering, specialising in EV systems. Experienced in mechanical design, vehicle dynamics and simulation-driven analysis using SOLIDWORKS, ANSYS Fluent, CFD/FEA and MATLAB/Simulink. As a Formula Student team member with Team Bedfordshire Racing, took chassis and aerodynamic components from concept through CAD design into workshop manufacture and assembly, working to weight, packaging, cost and project deadlines. Completed a CAE research internship on battery thermal management systems for EVs. Seeking a UK graduate engineering role, with particular interest in simulation, manufacturing systems and quality assurance.

KEY SKILLS

CAD & Design: SOLIDWORKS, 3D CAD modelling, design for manufacture, component packaging, technical drawings

Simulation & Analysis: ANSYS Fluent, SpaceClaim, CAE, Computational Fluid Dynamics (CFD), FEA, thermal analysis, mesh independence studies, turbulence modelling, design optimisation

Engineering Tools: MATLAB, Simulink, Microsoft Excel, Microsoft Word, PowerPoint

EV & Vehicle Systems: Battery systems, battery thermal management (BTMS), power electronics, motor control, automotive electronics, control systems, vehicle dynamics, thermodynamics and heat transfer

Manufacturing: Welding, metal cutting, chassis fabrication, vehicle assembly and component fitment, workshop health & safety, quality assurance of components

Professional: Cross-functional teamwork, technical report writing, problem solving, root cause investigation, working to regulations and deadlines

EDUCATION

MSc Automotive Engineering for Electric Vehicles — 2:1

May 2025 – May 2026

University of Bedfordshire, Luton, United Kingdom

Key modules: Computational Fluid Dynamics; Thermodynamics, Heat Transfer & Fluid Dynamics; Vehicle Dynamics & Performance; Automotive Electronics & Control Systems; Engineering Analysis & Numerical Modelling.

BEng (Hons) Electrical & Electronics Engineering (Electric Vehicles) — First Class

Jun 2020 – May 2024

Bharath Institute of Higher Education and Research, India

Key modules: Electrical Machines & Power Electronics; Electrical Circuits & Power Systems; Control Systems & Embedded Systems; Automotive Electronics; Electric Vehicle Technologies & Battery Systems.

ENGINEERING EXPERIENCE

Formula Student Team Member — Team Bedfordshire Racing, Luton, UK

Sep 2025 – Jun 2026

Class II vehicle project — chassis design, fabrication and quality assurance

- Designed and fabricated **Formula Student race car chassis components**, combining CAD-based design in SOLIDWORKS with hands-on workshop manufacture.
- Carried out metal cutting, welding, chassis fabrication, assembly and component fitment, delivering parts that met packaging and weight targets.
- Supported 3D CAD modelling and aerodynamic evaluation, feeding simulation results into performance-driven design improvements.
- Performed quality checks on manufactured components against drawings and Formula Student technical regulations, identifying and correcting build issues before assembly.
- Collaborated within a **24-member multi-disciplinary team** across design, manufacturing and vehicle build, coordinating work to project milestones and the competition timeline.
- Saved **over £500 against the build budget** through cost-conscious material and component decisions, working within a fixed team budget.
- Worked to workshop safety procedures and risk assessments throughout the vehicle development process.

CAE Engineer Intern (Research Internship) — Simulation Lab, Product Development Centre Nov 2024 – Jan 2025

Remote research internship — Battery Thermal Management Systems (BTMS) using nanofluids for EVs

- Completed a two-month CAE research internship investigating **battery thermal management systems (BTMS) using nanofluids** for electric vehicle applications.
- Analysed coolant and heat transfer behaviour for EV battery packs, applying thermal and fluid dynamics principles to a live product development research problem.
- Produced and presented research findings on nanofluid coolant performance, linking simulation-based analysis to practical EV thermal management design decisions.

ENGINEERING PROJECTS

MSc Dissertation — CFD Optimisation of a Multi-Element Front Wing

Completed May 2026

University of Bedfordshire | ANSYS Fluent, SpaceClaim, SOLIDWORKS

- Achieved an approximate **45% reduction in aerodynamic drag** while improving aerodynamic efficiency, through iterative design optimisation of a multi-element Formula Student front wing.
- Built and refined front wing geometries in CAD and quantified the effect of aerodynamic parameters on downforce, drag and overall vehicle performance.
- Ran mesh independence studies, selected appropriate turbulence models and analysed pressure distribution and velocity contours to validate results.
- Translated CFD results into engineering recommendations and produced a full technical report covering methodology, results and design justification.

TRAINING & PROFESSIONAL DEVELOPMENT

Artificial Intelligence programme: IIT Kanpur (1Stop), Jun – Aug 2021 — two-month programme, certificate of participation.